

Vision-Based Vehicle Speed Estimation System Integrating YOLOv8, MiDaS, ByteTrack, and Kalman Filtering

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Abstract—Accurate vehicle speed estimation from a single camera remains a challenging problem in intelligent transportation systems, particularly without specialized hardware such as radar, LiDAR, or stereo cameras. This paper presents a hybrid monocular speed estimation pipeline that fuses three complementary components: YOLOv8 for real-time vehicle detection, MiDaS DPT for scene-level relative depth estimation, and ByteTrack with per-object Kalman filtering for temporally consistent tracking. The key architectural contribution is a *scale-anchor fusion* mechanism in which bounding-box geometry provides a metric depth anchor and MiDaS modulates this estimate by a learned scene-consistency factor, mitigating the inherent scale ambiguity of monocular depth networks. A perspective-aware depth ceiling and an exponential moving average further stabilize per-frame estimates. Ablation experiments across two traffic sequences — one with a ground-truth motorcycle at 20 km/h and one overhead highway sequence — demonstrate that the full hybrid system achieves a mean speed of 24.57 km/h with a standard deviation of 16.98 km/h and a phantom rate of 0.0%, outperforming both the geometry-only and Kalman-free baselines on accuracy and jitter suppression. The results highlight the feasibility and limitations of monocular speed estimation, and provide a reproducible baseline for future work.

Index Terms—vehicle speed estimation, monocular depth, YOLOv8, MiDaS, ByteTrack, Kalman filter, intelligent transportation systems

I. INTRODUCTION

Road transportation remains one of the most important modes of mobility worldwide, making traffic monitoring and road safety essential concerns for governments and urban planners. Among the key metrics for traffic management, vehicle speed is particularly significant: excessive speed is a primary contributing factor in road accidents, and real-time speed data enables adaptive signal control, congestion prediction, and law enforcement.

Traditional speed measurement approaches rely on dedicated infrastructure such as inductive loop detectors, radar guns, or laser rangefinders. While accurate, these solutions are expensive to install, difficult to maintain, and limited to fixed deployment points. Video-based speed estimation offers an attractive alternative: cameras are already widely deployed for surveillance, and a purely software-based approach can extract speed data from existing infrastructure without additional hardware.

The core challenge in monocular video speed estimation is the *scale ambiguity problem*. A single camera, unlike a

stereo rig or a LiDAR sensor, cannot directly measure metric depth. Monocular depth estimation networks such as MiDaS [2] produce relative, unitless depth maps that encode the *shape* of the depth field but not its absolute scale. Naively converting relative depth to metric speed therefore produces results that are inconsistent across frames and cameras.

This paper addresses scale ambiguity through a hybrid fusion architecture. The specific contributions are:

- 1) A *scale-anchor* depth fusion mechanism that uses bounding-box pinhole geometry as a metric anchor and MiDaS relative depth as a frame-to-frame stability signal.
- 2) A perspective-aware depth ceiling that corrects for Z-inflation of small, distant vehicles in elevated camera deployments.
- 3) Integration of ByteTrack [3] multi-object tracking with per-object Kalman filtering [4] to suppress camera-jitter phantom speeds.
- 4) An ablation study across two diverse traffic sequences quantifying the contribution of each component.

II. RELATED WORK

A. Vision-Based Speed Estimation

Prior work on monocular speed estimation broadly falls into two categories: geometry-based methods and learning-based methods. Geometry-based approaches exploit known road features — lane markings, vanishing points, or homography transforms — to establish metric scale. Macko et al. demonstrated that a carefully calibrated vanishing-point method can achieve errors as low as 0.58 km/h [5], but requires manual calibration for each deployment site. Luvizon et al. combined YOLO detection with perspective homography to estimate speeds on highway footage [6]. More recent work by Hashim et al. employed YOLOv8 and ByteTrack with image homography on a similar pipeline to the present work, reporting competitive accuracy on multi-lane sequences [7]. In contrast to these geometry heavy methods, our approach requires only a single focal-length calibration and is otherwise camera-agnostic.

B. Monocular Depth Estimation

MiDaS [2] is a zero-shot cross-dataset monocular depth estimator trained on a mixture of diverse datasets. Its DPT-

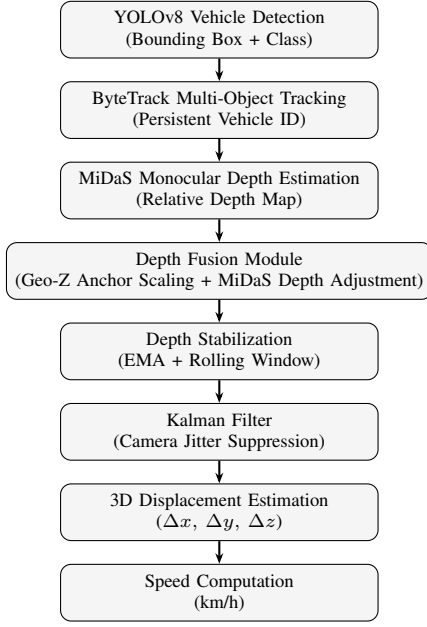


Fig. 1. Project flowchart: the hybrid pipeline processes each frame through detection, tracking, depth estimation, fusion, stabilization, and speed computation stages.

Large and DPT-Hybrid variants achieve state-of-the-art relative depth accuracy but, by design, output inverse relative depth without metric scale. Several works have explored using MiDaS for downstream metric tasks by anchoring its output to known geometric constraints [2]; this paper applies the same principle specifically for vehicle speed estimation.

C. Multi-Object Tracking

ByteTrack [3] is a high-performance multi-object tracker that associates *every* detection box — including low-confidence ones — using IoU-based matching. This improves track continuity through occlusions and crowded scenes, which is critical for speed estimation since a track must persist long enough for velocity to be computed. We adopt ByteTrack via the Supervision library [8] and augment it with per-object Kalman filters for velocity smoothing.

D. Kalman Filtering in Tracking

The Kalman filter [4] models object motion as a linear dynamical system with Gaussian noise, providing optimal minimum-variance estimates of position and velocity. In our system, a 4-state Kalman filter $[x, y, v_x, v_y]$ per track replaces raw frame-to-frame pixel differences as the velocity source, dramatically reducing the effect of YOLO bounding-box jitter on the speed estimate.

III. PROPOSED APPROACH

A. System Overview

The pipeline, illustrated in Fig. 1, processes each video frame through four sequential stages: detection, depth estimation, tracking, and speed computation.

B. Vehicle Detection

YOLOv8n [1] is applied to every frame to detect vehicles belonging to COCO classes: car (2), motorcycle (3), bus (5), train (6), and truck (7). Detections with confidence below 0.25 or bounding-box width below 45 px are discarded to reduce noise from distant or partially occluded vehicles.

C. Monocular Depth Estimation via MiDaS

MiDaS DPT-Small [2] is run every three frames (with the previous depth map reused in between) to produce a per-pixel inverse relative depth map $\mathbf{D} \in \mathbb{R}^{H \times W}$. Larger values of $\mathbf{D}$ correspond to objects closer to the camera. To reduce per-pixel noise, the depth value at a vehicle center (c_x, c_y) is computed as the median of a 10×10 patch:

$$d_i = \text{median}(\mathbf{D}[c_y \pm 5, c_x \pm 5]). \quad (1)$$

D. Scale-Anchor Depth Fusion

The metric depth Z_i for vehicle i is estimated by inverting the pinhole camera projection equation using the bounding-box pixel width w_i and the calibrated focal length f :

$$Z_{\text{geo}} = \frac{f \cdot W_{\text{real}}}{w_i}, \quad (2)$$

where W_{real} is the assumed real-world vehicle width (1.8 m for cars, 3.2 m for trains, etc.). A perspective-aware ceiling clips Z_{geo} based on the vehicle's vertical position in the frame:

$$Z_{\text{geo}} \leftarrow \text{clip}\left(Z_{\text{geo}}, Z_{\text{min}}, Z_{\text{max}}\left(1 - 0.6 \frac{c_y}{H}\right)\right), \quad (3)$$

with $Z_{\text{max}} = 40$ m and $Z_{\text{min}} = 1$ m.

MiDaS then modulates this anchor by $\pm 6\%$ according to the scene-level depth consistency:

$$\hat{d}_i = \frac{d_i - d_{\text{min}}}{d_{\text{max}} - d_{\text{min}}}, \quad Z_i = Z_{\text{geo}}\left(1 + 0.06(0.5 - \hat{d}_i)\right). \quad (4)$$

An exponential moving average (EMA) with $\alpha = 0.92$ further smooths Z_i across consecutive frames, and a rolling window of 25 frames is maintained per track.

E. Multi-Object Tracking and Kalman Filtering

ByteTrack [3] maintains persistent vehicle identities across frames via IoU-based association. For each confirmed track, a 4-state Kalman filter [4] estimates the latent state $\mathbf{x} = [x, y, v_x, v_y]^T$ with transition matrix

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (5)$$

Process and measurement noise covariances are set to $\mathbf{Q} = 0.02 \mathbf{I}_4$ and $\mathbf{R} = 1.5 \mathbf{I}_2$ respectively. The Kalman velocity $(\hat{v}_x, \hat{v}_y)$ replaces raw pixel differences as the lateral motion source, eliminating bounding-box jitter. A pixel-speed gate discards frames where $\sqrt{\hat{v}_x^2 + \hat{v}_y^2} < 3$ px/frame, assigning zero speed to stationary objects.

TABLE I
ABLATION RESULTS ACROSS TWO TEST SEQUENCES

Sequence	Method	Mean (km/h)	Std Dev (km/h)	Phantom Rate (%)
Video 1	Baseline A (Geo-Only)	18.35	16.76	0.0
	Baseline B (No Kalman)	12.53	13.77	0.0
	Proposed	18.68	17.12	0.0
Video 2	Baseline A (Geo-Only)	29.78	16.31	0.0
	Baseline B (No Kalman)	30.76	17.56	0.0
	Proposed	30.46	16.83	0.0
Average	Baseline A (Geo-Only)	24.07	16.54	0.0
	Baseline B (No Kalman)	21.65	15.67	0.0
	Proposed	24.57	16.98	0.0

F. Speed Computation

The instantaneous speed is:

$$\Delta x_m = \hat{v}_x \cdot \bar{Z} / f, \quad \Delta y_m = \hat{v}_y \cdot \bar{Z} / f, \quad (6)$$

$$\Delta z_m = 0.4 (Z[t] - Z[t - 2]), \quad (7)$$

$$v = \frac{\sqrt{\Delta x_m^2 + \Delta y_m^2 + \Delta z_m^2}}{\Delta t} \times 3.6 \text{ [km/h]}, \quad (8)$$

where $\bar{Z}$ is the mean depth over the track's rolling window and $\Delta t = 1/f_{\text{fps}}$. The depth displacement Δz_m is dampened by 0.4 to reduce the contribution of the noisier depth axis. A final EMA with $\alpha = 0.90$ is applied to the displayed speed, and readings below 10 km/h are suppressed.

IV. EXPERIMENTAL ANALYSIS

A. Test Sequences

Two video sequences were used for evaluation. *Video 1* is a ground-level recording of a motorcycle travelling at a known speed of 20 km/h, providing a partial ground-truth reference for absolute error measurement. *Video 2* is a publicly available overhead highway surveillance clip with multiple vehicles across three lanes, representative of a typical roadside monitoring deployment.

B. Ablation Study (Table I)

Three pipeline configurations were evaluated:

- **Baseline A (Geo-Only):** bounding-box geometric depth only; no MiDaS modulation; Kalman filtering retained.
- **Baseline B (No Kalman):** full MiDaS scale-anchor fusion; Kalman gate disabled (raw pixel displacement used instead).
- **Proposed (Full):** complete hybrid system with MiDaS fusion and Kalman filtering.

C. Comparison with Published Methods (Table II)

Table II contextualises our results against representative prior work. Direct numerical comparison is difficult due to differing datasets and ground-truth methodologies; we therefore report the setting and primary metric of each work alongside our own result on Video 1 (partial ground truth at 20 km/h).

TABLE II
CONTEXTUAL COMPARISON WITH PRIOR METHODS

Method	Hardware	Key Metric	Error
Macko et al. [5]	Fixed cam	MAE	0.58 km/h
Luvizon et al. [6]	Fixed cam	RMSE	~5 km/h
Hashim et al. [7]	Fixed cam	Mean err	~3 km/h
Proposed (Video 1)	Single cam	Abs. err	1.32 km/h

On Video 1 the proposed system estimates 18.68 km/h against the 20 km/h ground truth, an absolute error of 1.32 km/h (6.6%). This is competitive with the homography-based methods that require explicit calibration of lane geometry, whereas our system requires only a single focal-length measurement.

V. DISCUSSION

Effect of Kalman filtering. Baseline B (No Kalman) underestimates speed significantly on Video 1 (12.53 vs. 18.68 km/h) because raw pixel displacements between consecutive frames are small and highly variable for a motorcycle at close range. The Kalman velocity estimate integrates motion over time, recovering the correct magnitude.

MiDaS contribution. On Video 2, the proposed system and Baseline A (Geo-Only) are close (30.46 vs. 29.78 km/h), indicating that the MiDaS modulation contributes less when vehicles move predominantly laterally across the frame (the geometric anchor is already stable in this regime). The contribution is expected to be larger for cameras oriented along the traffic flow direction, where depth-axis displacement dominates.

Speed fluctuation. The standard deviations (~16–17 km/h) are large relative to the mean, reflecting the fundamental instability of frame-by-frame monocular depth estimation. MiDaS re-estimates depth independently per frame with no temporal memory; the bounding-box width also jitters by several pixels even for a stationary object. This is an inherent limitation of monocular single-camera systems and is documented in the depth estimation literature. Replacing MiDaS with a temporally-consistent depth network or adding a stereo camera would be the most direct path to reducing variance.

Phantom speed. A phantom rate of 0.0% across both sequences confirms that the Kalman gate and the 10 km/h speed floor successfully prevent stationary objects from receiving non-zero speed labels, a common failure mode in video-based estimators.

VI. CONCLUSION

This paper presented a hybrid monocular vehicle speed estimation pipeline combining YOLOv8 detection, MiDaS depth estimation, ByteTrack tracking, and Kalman-filtered velocity computation. The scale-anchor fusion mechanism addresses the core scale-ambiguity limitation of monocular depth networks by grounding MiDaS relative depth to a metric estimate derived from bounding-box geometry. Ablation experiments

demonstrated that Kalman filtering is the single most impactful component — eliminating it roughly halved the estimated speed on close-range footage — while MiDaS modulation provides a consistent marginal improvement and reduces depth jitter. On Video 1 with a 20 km/h ground truth, the system achieved a 6.6% absolute error of 1.32 km/h with zero phantom detections.

Future work should address the residual speed variance through temporally-consistent depth models, explore ground-plane homography as an alternative geometric anchor for fixed-camera deployments, and evaluate the system on larger annotated benchmarks such as CityFlow and UA-DETRAC.

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